

VAPO: Value-Model-Based RL for Advanced Reasoning

Achieves 60.4 on AIME 2024

ByteDance Seed | Base Model: Qwen2.5-32B

Why Value Models? Finer Credit Assignment

- **Value-Model-Free (e.g., GRPO, DAPO)**
 - Assign trajectory-level advantage uniformly across all tokens.
 - Less precise for multi-step reasoning where subtle errors cause failure.
- **Value-Model-Based (VAPO)**
 - Enable token-level credit assignment, tracing each action's impact.
 - Provide lower-variance estimates per token compared to Monte Carlo.
 - Better generalization across samples, improving

Three Challenges in Value-Model-Based RL

1. Value Model Bias

Caused by long trajectories and bootstrapped learning. Initial bias is high before RL.

2. Heterogeneous Lengths

Variable sequence lengths require different bias-variance tradeoffs in generalized advantage estimation.

3. Reward Signal Sparsity

Exacerbated by long Chain-of-Thought (CoT) reasoning; only the final <eos> token receives a reward.

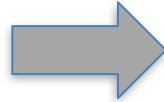
VAPO Framework: Integrated Solutions

Challenge 1: Value Model Bias



- Value-Pretraining
- Decoupled-GAE

Challenge 2: Heterogeneous Lengths



- Length-Adaptive GAE

Challenge 3: Reward Sparsity

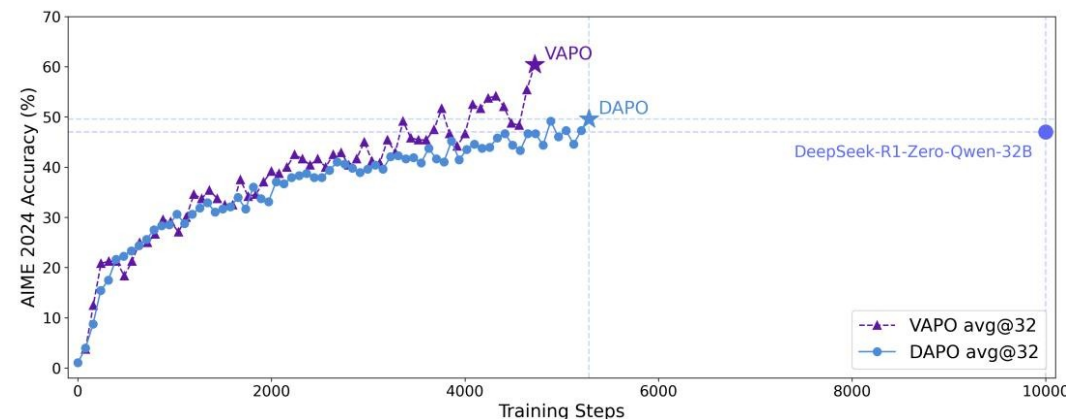
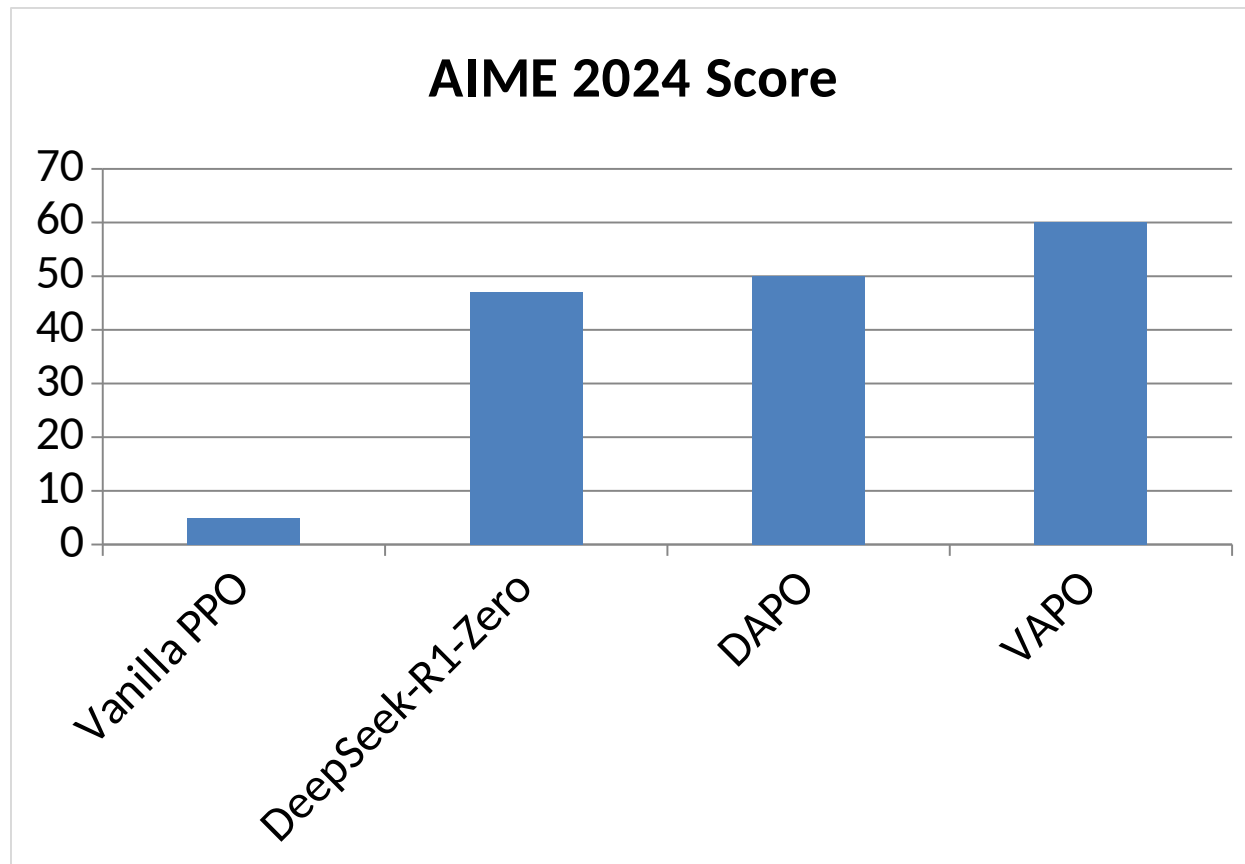


- Clip-Higher
- Token-level Loss
- Positive Example LM Loss
- Group-Sampling

Length-Adaptive GAE: Dynamic λ Adjustment

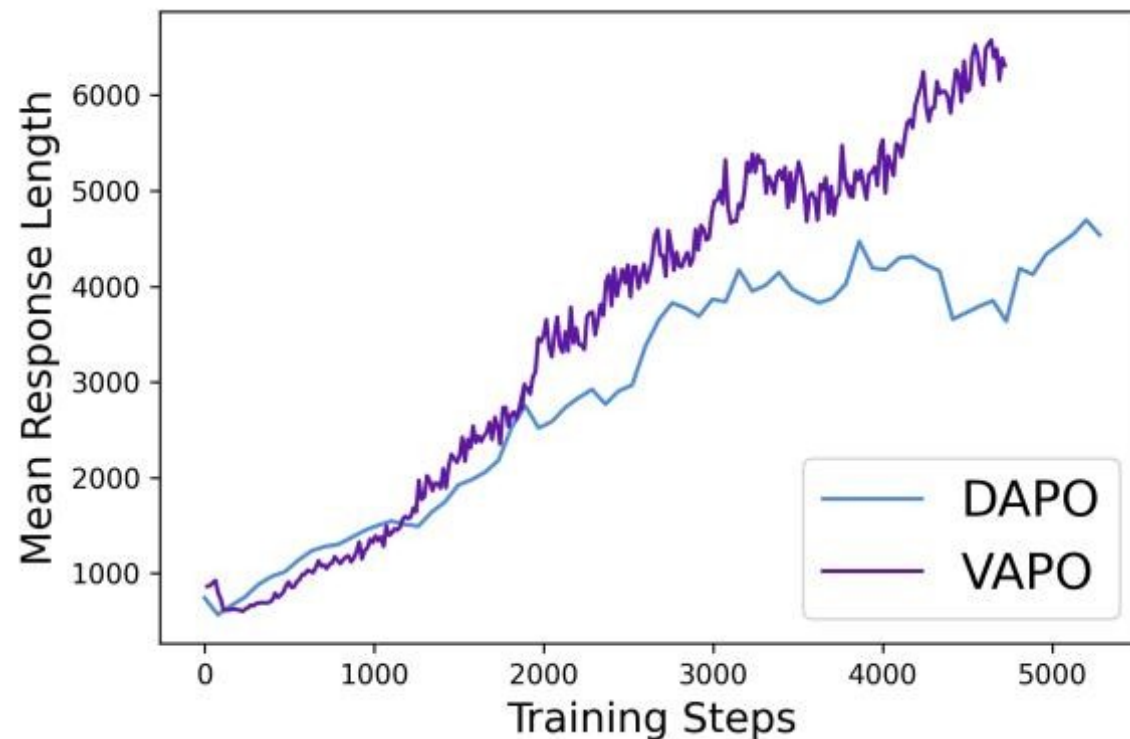
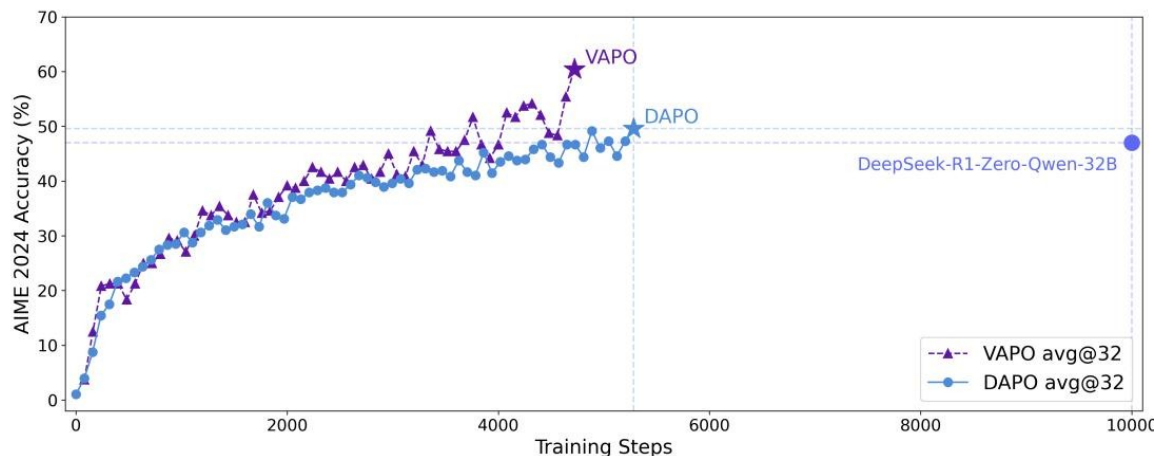
- A novel technique that dynamically adjusts the λ parameter in GAE based on response length.
- **Short Responses:**
 - Use lower λ (prioritizes lower variance).
- **Long Responses:**
 - Use higher λ (prioritizes lower bias).
- **Impact:**
 - Optimizes the bias-variance tradeoff across heterogeneous lengths unique to long-CoT reasoning.

VAPO Scores 60.4 on AIME 2024 (10+ Points Above SOTA)



Training curves showing VAPO consistently outperforming DAPO

Training Efficiency: SOTA in 5K Steps



- Reaches SOTA in ~5,000 steps (half the training budget of DAPO) with zero training crashes.
- Generates longer reasoning chains (~6,000 tokens vs ~4,500), indicating more elaborate problem-solving.

Ablation: Every Component Matters

Model Configuration	AIME24 Score	Drop from Full
VAPO (Full Framework)	60	-
w/o Group-Sampling	55	-5
w/o Positive Example LM Loss	54	-6
w/o Token-level Loss	53	-7
w/o Clip-Higher	46	-14
w/o Length-adaptive GAE	45	-15
w/o Decoupled-GAE	33	-27
w/o Value-Pretraining	11	-49

Key Takeaways: Value-Model-Based RL Is Back

- Value-model-based methods can surpass value-free RL when core challenges (bias, lengths, sparsity) are addressed.
- Value-Pretraining is the single most important component — without it, the score drops from 60 to 11.
- Length-adaptive GAE is critical for handling the heterogeneous sequence lengths unique to long-CoT reasoning.
- VAPO is highly stable and training-efficient (zero crashes, reaches SOTA in 5,000 steps).
- Implication: Value-model-based RL, paired with proper engineering, is a powerful paradigm for future reasoning models.